

# Linear Algebra Recitation

## Week 3

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# The subspace of a vector space

- **Definition.** A set  $V$  is a vector space iff it is equipped with two operations, vec. add & scalar mul,  $u, v, w \in V$   
 $c_1, c_2 \in \mathbb{R}$

- $V$  is closed under vec. add scalar mul  
 $\forall v, u \in V \uparrow \quad \forall c \in \mathbb{R} \uparrow$   
 $\forall u \in V \quad \downarrow \quad \downarrow$   
 $cu \in V$
- $\vec{0} \in V$

- **Definition.** A set  $S$  is a subspace of a vector space  $V$  iff

- $S$  is a subset of  $V$ .
- $S$  is a vector space (with same operations as  $V$ )

- Note that the operations of  $S$  are "inherited" from  $V$ .

- ①  $v + u = u + v$
- ②  $u + (v + w) = (u + v) + w$
- ③ Vector addition must have unique identity element (zero vector)
- ④ every vector  $v$  in vector space must have unique inverse element ( $v^{-1}$ )
- ⑤  $1 \cdot v = v$
- ⑥  $c_1(c_2 v) = (c_1 c_2) v$
- ⑦  $c(v + u) = cv + cu$
- ⑧  $(c_1 + c_2) v = c_1 v + c_2 v$

$$v + \frac{v^{-1}}{-v} = e$$

$$c \cdot (x, y) = (cx, cy)$$

$$c \star (x, y) = (cx, y)$$

$$1 \star (x, y) = (x, y)$$

$$c_2 \star (c_1 \star (x, y)) = c_2 \star (c_1 x, y)$$

$$= (c_2 c_1 x, y)$$

$$= (c_2 c_1) \star (x, y)$$

~~$$\textcircled{1} (x_1, y_1) \star (x_2, y_2) = (x_1 x_2, y_1 y_2)$$~~

~~$$(x_2, y_2) \star (x_1, y_1) = (x_2 x_1, y_2 y_1)$$~~

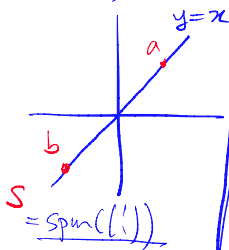
~~$$\textcircled{3} (x, y) \star e = (x, y)$$~~

~~$$e = (1, 1)$$~~

~~$$\textcircled{4} (x, y) \star \frac{(x, y)^T}{(\frac{1}{x}, \frac{1}{y})} = e = (1, 1)$$~~

# The subspace of a vector space: Exercises

- Is  $S_1 = \{(x, y) \in \mathbb{R}^2 : x - y = 0\}$  a subspace of  $\mathbb{R}^2$ ?



①  $\checkmark (0, 0) \in S$

②  $a = (x_1, y_1) \quad a + b \in S$

$b = (x_2, y_2) \quad a + b = (x_1 + x_2, y_1 + y_2)$

$x_1 - y_1 = 0$

$x_2 - y_2 = 0$

$x_1 + x_2 - (y_1 + y_2) = 0 ?$

$x_1 - y_1 + x_2 - y_2 = 0$

③  $c \in \mathbb{R}$

$ca = (cx_1, cy_1) \in S \quad cx_1 - cy_1 = 0$

- Is  $S_2 = \{(x, y) \in \mathbb{R}^2 : x - y = 1\}$  a subspace of  $\mathbb{R}^2$ ?

$a + b \in S$

$(cx_1 + x_2, cy_1 + y_2)$

# The subspace of a vector space: Exercises

- Is  $\mathcal{P}_n$ , the set of real polynomials of degree at most  $n$ , a subspace of  $\mathbb{R}[x]$ , the vector space of all real polynomials? First, show that  $\mathbb{R}[x]$  is a vector space (infinite dimensional).

$\mathbb{R}[x]$  a vector space?

①  $f(x) \equiv 0 \in \text{zero vector}$

②  $p(x), r(x) \in \mathbb{R}[x]$

$$p(x) + r(x) = r(x) \in \mathbb{R}[x]$$

③  $c p(x) \in \mathbb{R}[x]$

- Is  $S = \{p(x) \in \mathcal{P}_n : p(1) = 0\}$  a subspace of  $\mathcal{P}_n$ ?

$$p(x), r(x) \in \mathcal{P}_n$$

$$\textcircled{1} r(x) = p(x) + r(x)$$

$$r(1) = p(1) + r(1) = 0$$

$$\textcircled{2} c p(x), c p(1) = 0$$

$$\textcircled{1} f(x) \equiv 0 \in \mathcal{P}_n$$

$$\textcircled{2} s(x), w(x) \in \mathcal{P}_n$$

$$s(x) + w(x) \in \mathcal{P}_n$$

$$\textcircled{3} c s(x) \in \mathcal{P}_n$$

$$\textcircled{3} f(x) \equiv 0 \in S$$

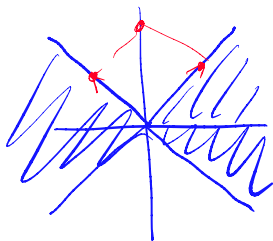
# The subspace of a vector space: Exercises

- Is  $S = \{p(x) \in \mathcal{P}_n : p(1) = 1\}$  a subspace of  $\mathcal{P}_n$ ?

$$r(x) = \underbrace{p(x)} + \underbrace{q(x)} = 2 \neq 1$$

$$c p(x) = c \neq 1$$

- Is  $S = \{(x, y) \in \mathbb{R}^2 : x^2 - y^2 \geq 0\}$  a subspace of  $\mathbb{R}^2$ ?



$$x^2 \geq y^2$$

$$\textcircled{1} x \geq 0, y \geq 0$$

1<sup>st</sup> quadrant

$$x \geq y$$

$$\textcircled{2} x \geq 0, y \leq 0$$

4<sup>th</sup> quadrant

$$x \geq -y$$

$$\textcircled{3} x \leq 0, y \geq 0$$

2<sup>nd</sup> quadrant

$$-x \geq y$$

$$\textcircled{4} x \leq 0, y \leq 0$$

3<sup>rd</sup> quadrant

$$-x \geq -y$$

# The subspace of a vector space: Exercises

- Is the set of functions  $f : \mathbb{R} \rightarrow \mathbb{R}$  that have a discontinuity at  $x = 1$  a vector space? *No.*

$$\left. \begin{aligned} f(x) &= \begin{cases} 0 & x \leq 1 \\ 1 & x > 1 \end{cases} \\ g(x) &= \begin{cases} 0 & x \leq 1 \\ -1 & x > 1 \end{cases} \end{aligned} \right) f(x) + g(x) = 0 \text{ for all } x \in \mathbb{R}. \\ \text{(Continuous at } x=1 \text{)}$$

- Is  $C^1$ , the set of once differentiable functions a vector space?

*Yes.*  $f, g \in C^1$

$$\Rightarrow (f+g)' = f' + g' \Rightarrow f+g \in C^1$$

$$(cf)' = cf' \Rightarrow cf \in C^1$$

# Subspaces of $\mathbb{R}^n$ related to a matrix

- The row and column space of a matrix. The dimension of either space is the **rank** of the matrix.  $\dim(\text{col}(A)) = \dim(\text{row}(A)) = \text{rank}(A)$ .

$$\overset{\parallel}{\dim(\text{col}(A^T))}$$

- The nullspace of a matrix (also called the kernel of a matrix). The dimension of the nullspace,  $\dim(\mathcal{N}(A))$ , is called the nullity of  $A$ .

**Definition.** a vector  $v \in V$  is in  $\mathcal{N}(A)$  iff  $Av = \vec{0}$

- Showing linear independence of vectors is related to finding the nullspace of a matrix.  $u, v, w$ , L.I.?

$$c_1 u + c_2 v + c_3 w = \vec{0}$$

if  $c_1 = c_2 = c_3 = 0 \Rightarrow u, v, w$  are L.I.

$$[u \mid v \mid w] \begin{bmatrix} c_1 \\ c_2 \\ c_3 \end{bmatrix} = 0$$

$$Av = 0$$

$$\begin{aligned} \mathcal{N}(A) &= \{ \vec{0} \} \\ \Rightarrow c_1 = c_2 = c_3 &= 0 \\ \Rightarrow u, v, w &\text{ L.I.} \end{aligned}$$

# Matrices as a linear transformation

- All matrices are linear transformations.

Conversely, if  $T : V \rightarrow W$  is a linear transformation and  $V, W$  are **finite dimensional** vector spaces, then there is a matrix representation for  $T$ .

**Definition:** A transformation  $T : V \rightarrow W$  is linear iff

$u, v \in V$  ✓

$$\begin{aligned} \textcircled{1} & T(u+v) = T(u) + T(v) \\ \textcircled{2} & T(cu) = cT(u) \end{aligned} \quad \Rightarrow \quad \begin{aligned} & T(cu+v) \\ & = cT(u) + T(v) \end{aligned} \quad \checkmark$$

- Consequences of linearity:

✱ if  $T$  is a linear transformation such that  $T\left(\begin{bmatrix} 1 \\ 2 \end{bmatrix}\right) = \begin{bmatrix} -2 \\ 1 \end{bmatrix}$  and

$T\left(\begin{bmatrix} -1 \\ 1 \end{bmatrix}\right) = \begin{bmatrix} 3 \\ -1 \end{bmatrix}$ , then find  $T\left(\begin{bmatrix} 0 \\ 1 \end{bmatrix}\right)$ .

$$\begin{aligned} T\left(\begin{bmatrix} 1 \\ 2 \end{bmatrix} + \begin{bmatrix} -1 \\ 1 \end{bmatrix}\right) &= T\left(\begin{bmatrix} 0 \\ 1 \end{bmatrix}\right) \\ &= T\left(\begin{bmatrix} 1 \\ 2 \end{bmatrix}\right) + T\left(\begin{bmatrix} -1 \\ 1 \end{bmatrix}\right) \\ &= \begin{bmatrix} -2 \\ 1 \end{bmatrix} + \begin{bmatrix} 3 \\ -1 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \end{aligned}$$

# Matrices as a linear transformation

$$\underline{Av} = \underline{w}$$

- Let  $A$  be an  $n \times m$  matrix. What is the domain and codomain of the matrix transform defined by  $T(v) = Av$ ?

$$\begin{matrix} n \\ \boxed{\begin{matrix} \sim \\ \sim \\ \sim \\ \sim \end{matrix}} \end{matrix} \begin{matrix} m \\ \boxed{\begin{matrix} \sim \\ \sim \end{matrix}} \end{matrix} \in \mathbb{R}^m = \begin{matrix} \boxed{\begin{matrix} \sim \\ \sim \end{matrix}} \end{matrix} \in \mathbb{R}^n \quad T: \mathbb{R}^m \rightarrow \mathbb{R}^n \quad \checkmark$$

- Let  $T: \mathbb{R}^2 \rightarrow \mathbb{R}^3$  be defined as  $\underline{T}\left(\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}\right) = \begin{bmatrix} x_1 + x_2 \\ x_2 \\ x_1 - x_2 \end{bmatrix}$ . Is there a YES matrix representation for this transformation?

$$A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \\ 1 & -1 \end{bmatrix}$$

- Let  $T: \mathbb{R}^3 \rightarrow \mathbb{R}^2$  be defined as  $T\left(\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}\right) = \begin{bmatrix} x_1 x_3 \\ \sqrt{x_2} \end{bmatrix}$ . Is there a NO matrix representation for this transformation?

$$A = \begin{bmatrix} \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} x_1 + x_3 \\ \sqrt{x_2} \end{bmatrix}$$

$$(1+2)$$

$$1 \cdot 3 + 2 \cdot 4 ?$$

NO

$$a = \begin{pmatrix} 2 \\ 2 \end{pmatrix}$$

$$b = \begin{pmatrix} 3 \\ 3 \end{pmatrix}$$

$$c = \begin{pmatrix} 4 \\ 4 \end{pmatrix}$$

$$d = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$